

New GEN Health AI

An AI-Powered Medical Consultation System

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Challenges In Current Medical AI

→ Generic Responses:

- Traditional chatbots give instant but vague answers
- Do not ask follow-up questions
- Lack of clinical context leads to inaccurate suggestions

Ex-A headache could be tension, migraine, meningitis, or a tumour.

Without knowing duration, severity, and accompanying symptoms, any diagnosis is essentially a guess.

→ Isolated Modalities:

- Text-based and image-based systems work separately
- Users need to switch between multiple tools
- No unified platform handles both together

→ Cross-Domain Errors:

- Images may be sent to the wrong model
- No proper domain identification step
- Leads to incorrect and unsafe predictions

Project Objectives

1 Clinical Interview AI

Build a conversational system that asks follow-up questions before diagnosing — replicating how real physicians conduct clinical interviews.

2 Domain-Aware Image Routing

Implement a multi-domain image classification pipeline that correctly identifies the image type before running any specialist CNN model.

3 RAG-Grounded Responses

Integrate retrieval-augmented generation so AI answers are grounded in real medical documents, reducing hallucinated information.

4 Responsive Frontend

Develop a clean, accessible React interface that makes the system usable for non-technical users including those in rural areas.

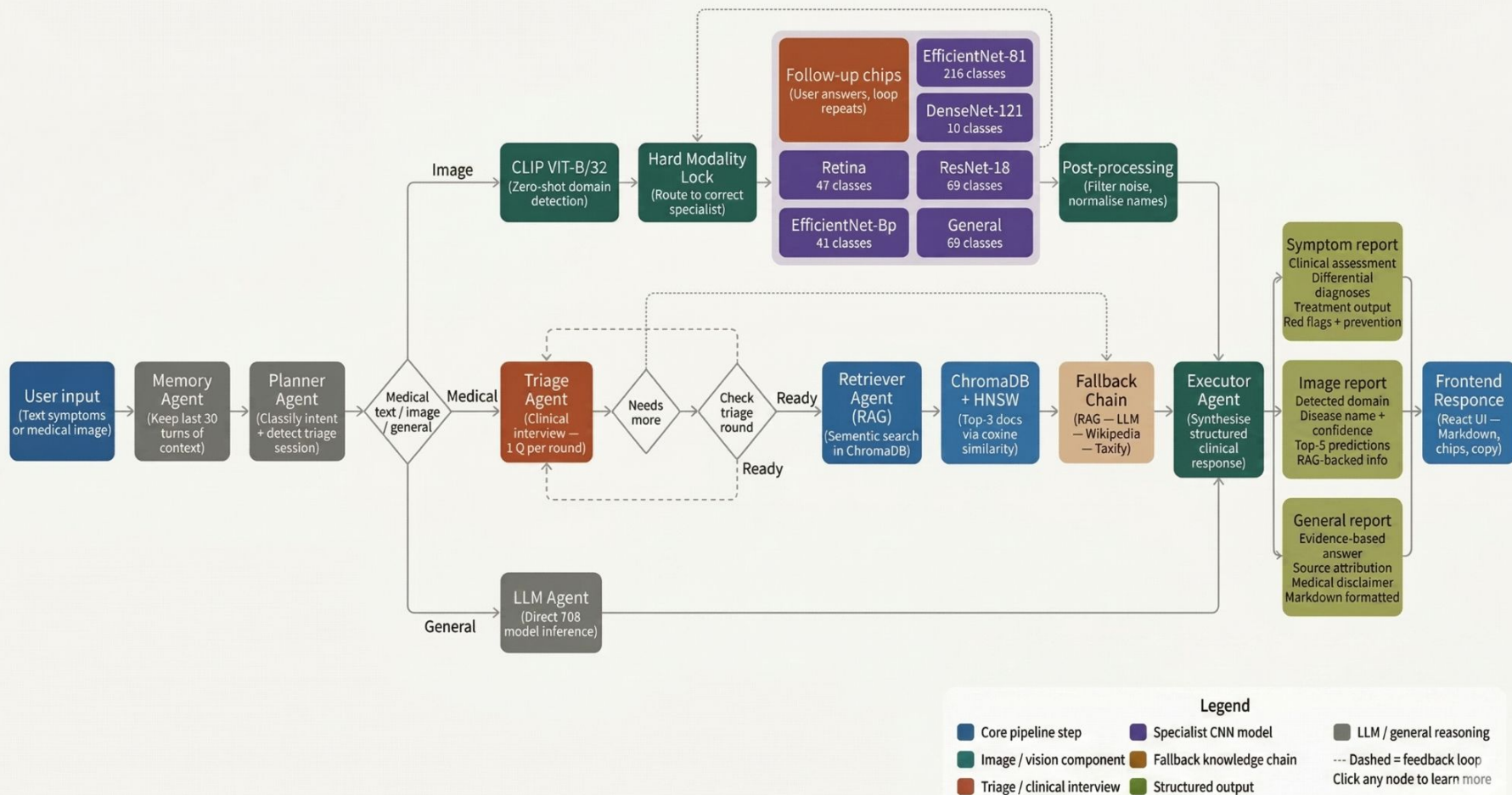
5 Multi-Modal Unification

Demonstrate how LLMs, CNNs, vision-language models, and knowledge retrieval can be combined into one coherent platform.

6 Stateful Conversations

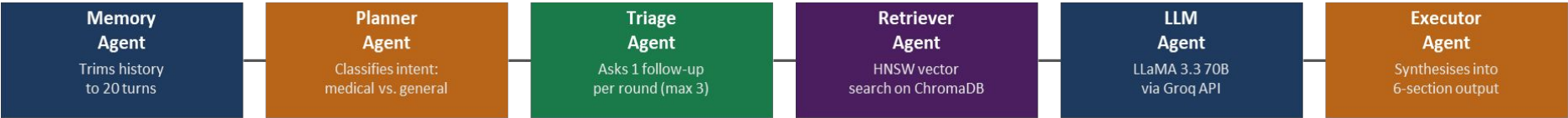
Maintain conversation history across multiple turns so patient details collected in early rounds improve later diagnosis quality.

System Workflow

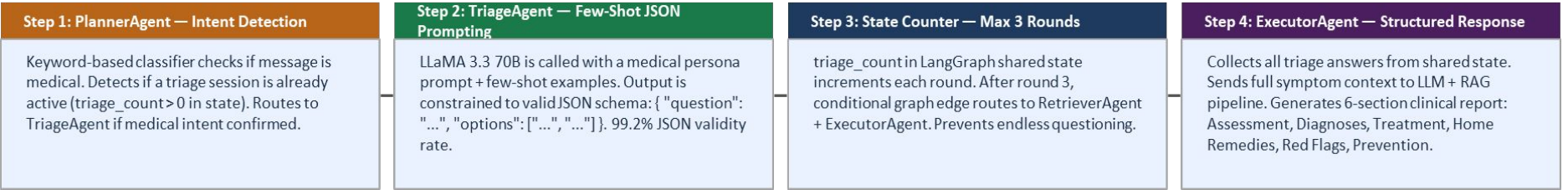


Clinical Triage Agent — How It Is Implemented

① LangGraph State Machine — Directed Acyclic Graph (DAG) with 9 Agents



② TriageAgent — Step-by-Step Implementation (How Each Round Works)



③ Key Technologies & Measured Performance

LLM	Framework	Prompting	Memory	Routing	Output
LLaMA 3.3 70B via Groq API	LangGraph 1.1.2 StateGraph DAG	Few-Shot + Schema Enforcement	Sliding Window 20-turn context	Conditional Graph Edges (LangGraph)	Structured JSON 6-section template

④ Triage Agent Performance Results (Section 8.2 — tested across 50 clinical cases)

Metric					
94.0%	99.2%	87.0%	91.0%	3.2 s	98.0%
Intent Classification (medical vs. general)	Structured JSON Output Validity	Differential Diagnosis Completeness	Red-Flag Symptom Detection	Avg. Response Generation Time	Out-of-Domain Query Rejection
Target: > 90%	Target: > 98%	Target: > 80%	Target: > 88%	Target: < 5 s	Target: > 96%

Clinical Triage Agent — Core Innovation

The Triage Agent transforms NewGenHealthAI from a passive chatbot into an active diagnostic interviewer — asking one follow-up question per round before forming a diagnosis, exactly like a real physician conducts a clinical interview.

Patient: "I have a headache"

AI — Round 1: "How long have you had this headache?"

< 24h

1–3 days

A week+

Chronic

Patient: Clicks: "1–3 days"

AI — Round 2: "How would you describe the pain?"

Mild/dull

Moderate

Severe/throbbing

Sharp

Patient: Clicks: "Severe/throbbing"

AI — Round 3: "Any other symptoms?"

Nausea

Light sensitivity

Fever

None

Final Output: Full clinical assessment — differential diagnoses, treatments, red flags, home remedies, prevention

Final Output Sections

1. Clinical Assessment

2. Possible Diagnoses (ranked)

3. Recommended Treatment

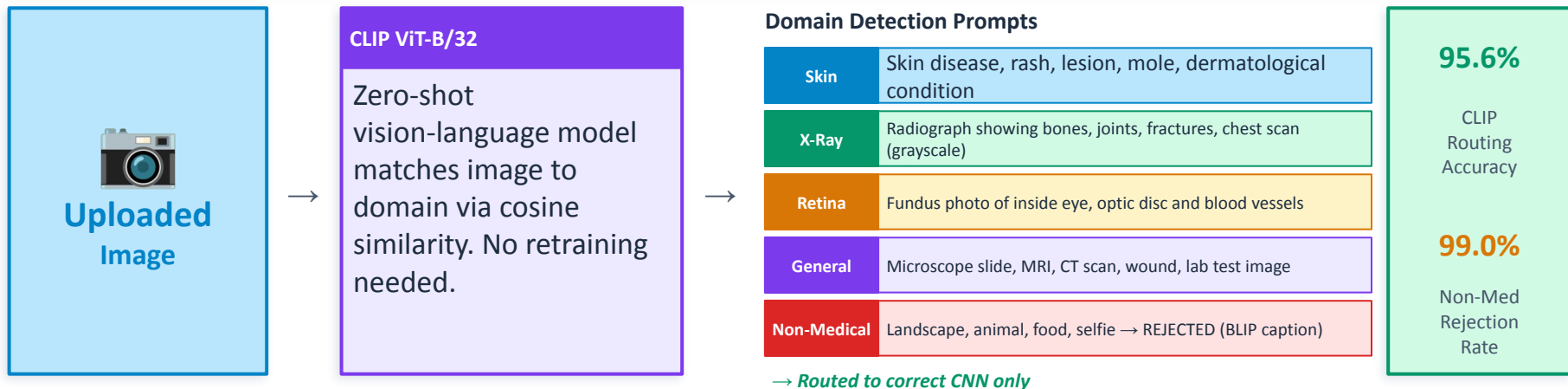
4. Home Remedies

5. When to See a Doctor

6. Prevention Tips

CLIP-Based Domain Routing — Hard Modality Lock

CLIP (OpenAI ViT-B/32) compares an uploaded medical image against five domain-specific text prompts using cosine similarity. The highest-matching domain wins — and only that specialist CNN processes the image. This prevents cross-domain classification errors entirely.



Hard Modality Lock — Why It Matters

Without a domain check, a skin photo sent to the X-ray model could be misidentified as a 'fracture', and a retinal image sent to the skin model might appear to have 'melanoma'. The Hard Modality Lock guarantees each image is only ever processed by its matching specialist model — completely eliminating cross-domain errors. Non-medical images (selfies, landscapes) are rejected with a BLIP-generated caption instead of a false medical diagnosis.

Multi-Model Image Classification — 4 Specialist CNNs

After CLIP routes the image to the correct domain, one of four specialist deep learning models performs disease classification. Together they cover 342 disease classes across four medical imaging domains.

Skin CNN — EfficientNet-B4-NS

87.4%

Disease Classes:	216	Sample: · Melanoma · Basal Cell Carcinoma · Squamous Cell Carcinoma
Input Size:	380×380px	· Dermatitis
Domain:	Dermoscopic skin diseases	· Psoriasis
Dataset:	ISIC 2020 Archive	
F1 Score:	86.3	

X-Ray CNN — DenseNet-121

91.2%

Disease Classes:	10	Sample: · Fractures (wrist, elbow, shoulder) · Pneumonia · Cardiomegaly · Pleural Effusion
Input Size:	224×224px	
Domain:	Musculoskeletal & Chest	
Dataset:	MURA + NIH Chest	
F1 Score:	90.1	

Retina CNN — EfficientNet-B0

83.6%

Disease Classes:	47	Sample: · Diabetic Retinopathy · Glaucoma · AMD · Macular Hole
Input Size:	224×224px	
Domain:	Retinal fundus diseases	
Dataset:	APTOS + RFMiD	
F1 Score:	82.1	

General CNN — ResNet-18

79.3%

Disease Classes:	69	Sample: · Histology slides · MRI lesions · CT anomalies · Wound types
Input Size:	224×224px	
Domain:	Mixed pathology images	
Dataset:	MedMNIST + Custom	
F1 Score:	77.8	

RAG Pipeline — Evidence-Based Medical Responses

Retrieval-Augmented Generation (RAG) improves AI responses by retrieving relevant medical knowledge before generating answers, ensuring accuracy and reducing hallucinations.

1 PDF Ingestion

- Medical PDFs are parsed and split into smaller text chunks
- Chunking preserves context for better retrieval

2 Embedding

- Each text chunk is converted into a numerical vector
- Uses MiniLM model for semantic understanding

3 ChromaDB Store

- All vectors stored in ChromaDB
- Uses HNSW index for fast similarity search

4 Query Encoding

- User query is converted into the same vector space
- Enables meaningful comparison with stored data

5 Top-3 Retrieval

- System retrieves the 3 most relevant chunks
- Based on cosine similarity search

6 LLM Synthesis

- Retrieved data + user query sent to LLM
- Generates accurate, evidence-based response

0.81

Precision@3

0.76

Recall@3

0.84

MRR

0.88

Faithfulness

0.85

Answer Relevance

18 ms

Search Latency

Agent Involved

Agent	Role	AI Technique
MemoryAgent	Keeps the conversation history trimmed to the last 20 turns to stay within LLM token limits	Sliding window context management
PlannerAgent	Classifies whether the input is a medical query or general question; detects active triage sessions	Keyword-based intent classification
TriageAgent	Asks one follow-up question per round to gather clinical details before forming a diagnosis	Structured JSON generation via few-shot prompting
RetrieverAgent	Searches the medical knowledge base using semantic similarity to find relevant documents	Dense retrieval with HNSW vector search
LLMAgent	Generates a direct LLM response when RAG retrieval is not sufficient	70B parameter instruction-tuned LLM
WikipediaAgent	Falls back to Wikipedia for general health information	Wikipedia API integration
TavilyAgent	Falls back to real-time web search for the most up-to-date medical information	Tavily AI search API
ExecutorAgent	Synthesises all gathered information into a structured, formatted clinical response	Abstractive summarisation and template generation
ExplanationAgent	Reserved for future explainability features (XAI)	Extensibility placeholder

AI Techniques Used

#	AI Technique	Where Applied in the System	Key Detail
1	Zero-Shot Classification	CLIP domain routing — identifies image type (skin, X-ray, retina, general, non-medical) with no domain-specific training	CLIP ViT-B/32 — 95.6% routing accuracy across 500 test images
2	Transfer Learning	All four CNN specialist models — fine-tuned from pre-trained ImageNet weights to medical domains	EfficientNet-B4-NS (87.4%), DenseNet-121 (91.2%), EfficientNet-B0 (83.6%), ResNet-18 (79.3%)
3	Contrastive Learning	CLIP ViT-B/32 — trained on 400 million image-text pairs to align image and language representations	149 million parameters; enables zero-shot matching of images to text descriptions
4	Vision Transformer (ViT)	CLIP image encoder — extracts global image features via self-attention across image patches	ViT-B/32 patch size; used for hard modality lock with 99% non-medical rejection
5	Dense Semantic Retrieval	RAG pipeline — converts queries & documents to 384-d vectors for cosine similarity search	all-MiniLM-L6-v2 embeddings; HNSW index in ChromaDB; 18 ms search latency
6	Retrieval-Augmented Generation (RAG)	Medical Q&A — LLM responses grounded in ~12,000 PDF text chunks from knowledge base	Faithfulness 0.88, Answer Relevance 0.85 (RAGAS); Precision@3 = 0.81
7	Multi-Agent Orchestration	LangGraph DAG with 9 specialised agents (Memory, Planner, Triage, Retriever, LLM, Wikipedia, Tavily, Executor, Explanation)	StateGraph with conditional routing; 94% intent classification accuracy
8	Structured Output Generation	TriageAgent — constrains LLM output to valid JSON for follow-up Q&A chips and diagnostic fields	99.2% JSON validity; few-shot prompting with schema enforcement
9	Sliding Window Context Mgmt.	MemoryAgent — trims conversation history to last 20 turns to stay within LLM token limits	Ensures LLaMA 3.3 70B context window is never exceeded during multi-round triage
10	HNSW Approximate NN Search	ChromaDB vector database — fast $O(\log n)$ search across 12,000 medical document chunks	Cosine distance metric; 18 ms average search latency; replaces brute-force kNN
11	NoisyStudent Self-Training	EfficientNet-B4-NS (Skin CNN) — semi-supervised training improves generalisation	Top-1 Acc 87.4% across 216 skin disease classes; best performing specialist CNN
12	Prompt Engineering	All LLM calls — medical persona, round-specific triage instructions, structured response templates	6-section output: Clinical Assessment, Diagnoses, Treatment, Home Remedies, Red Flags, Prevention

Future Scope & Conclusion

Future Scope:

- **Explainable AI (XAI):** Adding Grad-CAM visualizations to highlight diagnostic regions in images.
- **Local Deployment:** Quantizing models for offline use in rural medical clinics.
- **EHR Integration:** Building FHIR-compliant adapters for real-world hospital data.

Conclusion:

- Combines **text + image analysis** in one system
- Uses **LLM + CNN + RAG + Agents**
- Provides a **clinically-inspired AI consultation pipeline**